Extended ER Features

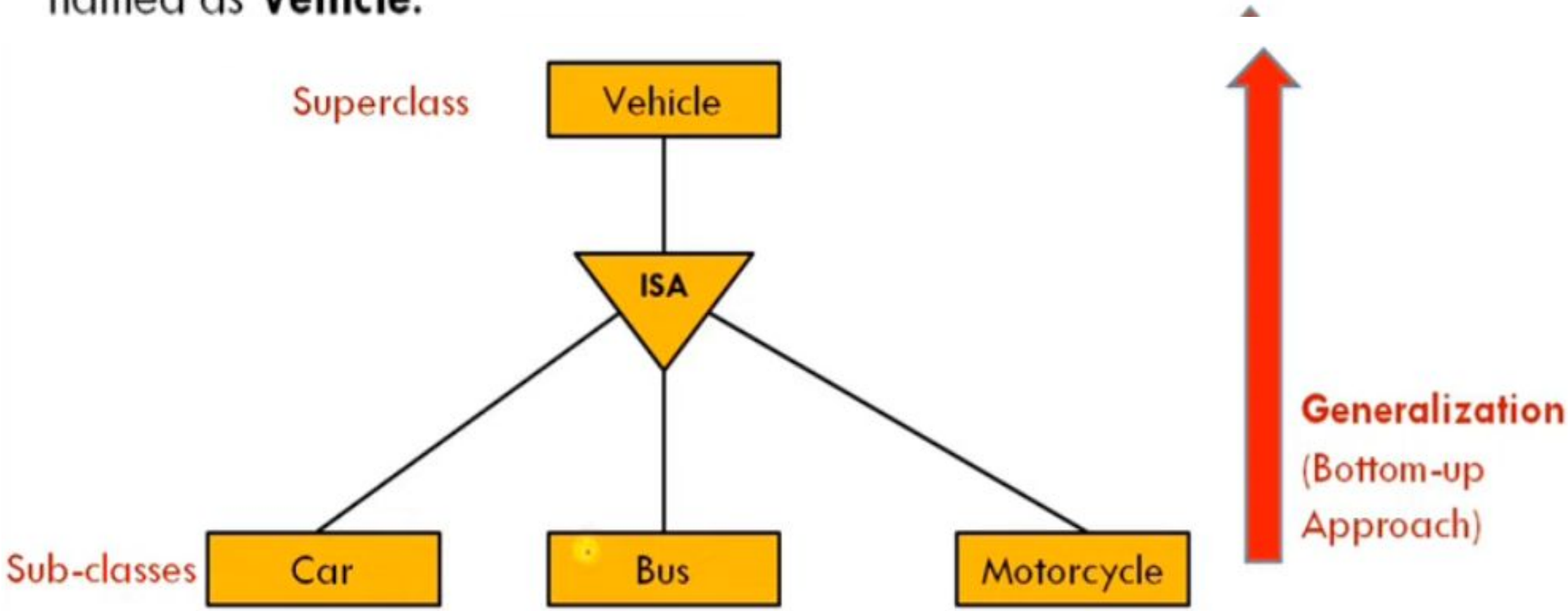
https://beginnersbook.com/2018/11/dbms-generalization/

Extended Entity Relationship Model

- Three new concepts added to existing ER model :
  1. Generalization
  2. Specialization
  3. Aggregation

- **Generalization** is the process of extracting common properties from a set of entities and create a generalized entity from it.
- **Generalization is a "bottom-up approach"** in which two or more entities can be combined to form a higher level entity if they have some attributes in common.
  - subclasses are combined to make a superclass.
- **Generalization is used** to emphasize the similarities among lower-level entity set and to hide differences in the schema

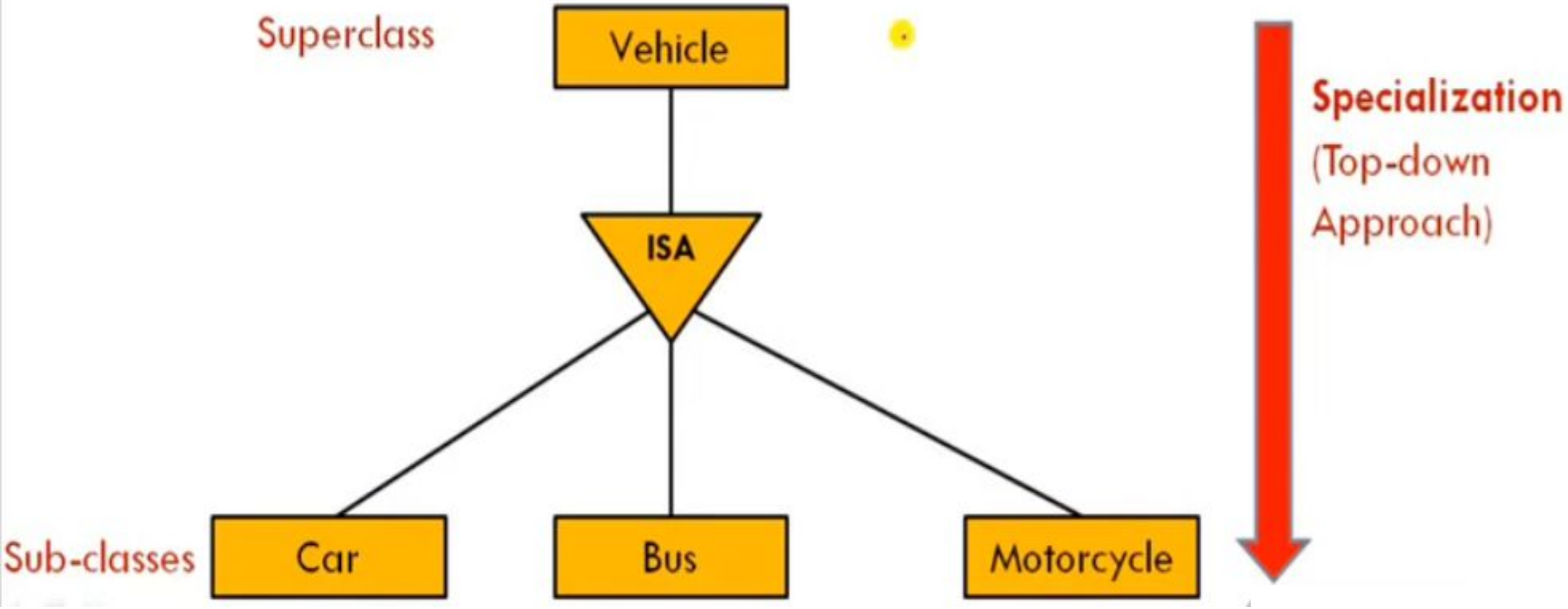
- Consider we have 3 sub entities Car, Bus and Motorcycle. Now these three entities can be generalized into one higher-level entity (or super class) named as **Vehicle.**



Specialization

- Specialization is **opposite** of Generalization
- In **Specialization**, **an entity is broken down into sub-entities based on their characteristics.**
- **Specialization is a "Top-down approach"** where higher level entity is specialized into two or more lower level entities.
- **Specialization is used** to identify the subset of an entity set that shares some distinguishing characteristics.
- **Specialization** can be repeatedly applied to refine the design of schema
- depicted by triangle component labeled **ISA**

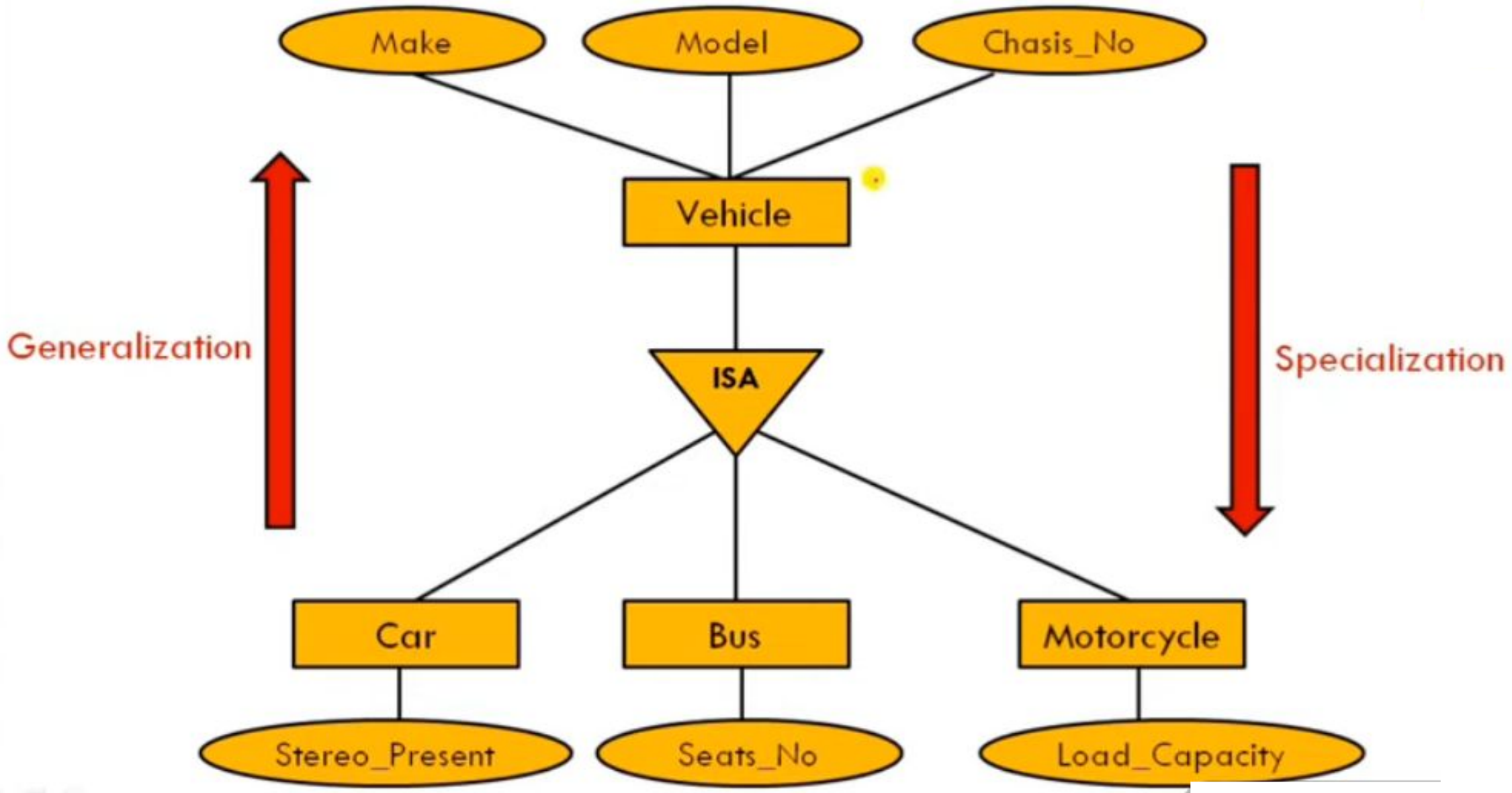
- **Vehicle** entity can be a Car, Truck or Motorcycle.
  - Normally, the superclass is defined first, the subclass and its related attributes are defined next, and relationship set are then added.



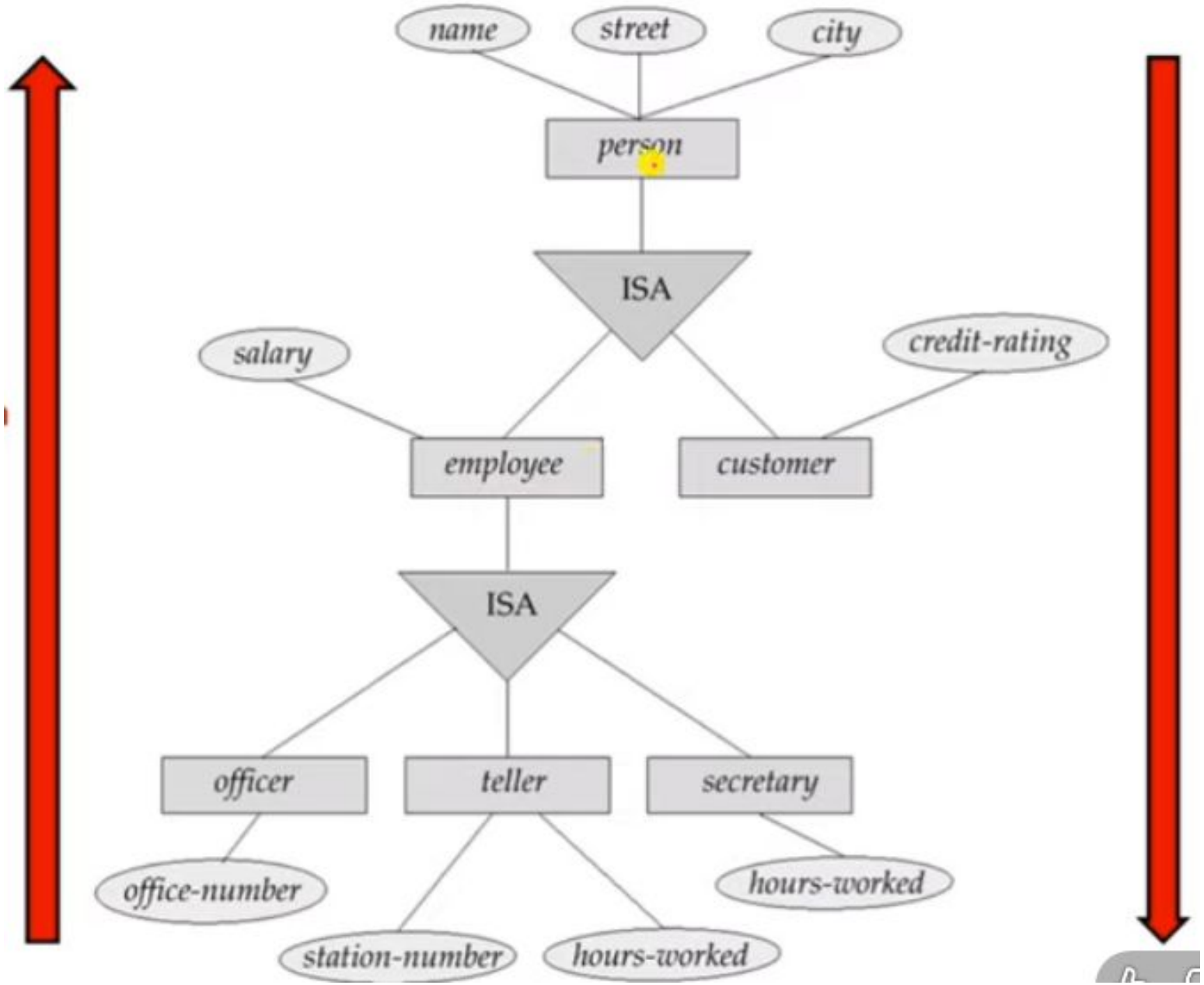
Inheritance

- Inheritance is an important feature of Generalization & Specialization
- Attribute inheritance allows lower level entities to inherit the attribute of higher level entities.

This also extends to **Participation Inheritance** in which ***relationships involving higher-level entity-sets are also inherited by lower-level entity-sets.***


Make
Model
Chasis_No
Vehicle
Generalization
Specialization
ISA
Car
Bus
Motorcycle
Stereo_Present
Seats_No
Load_Capacity

Generalization



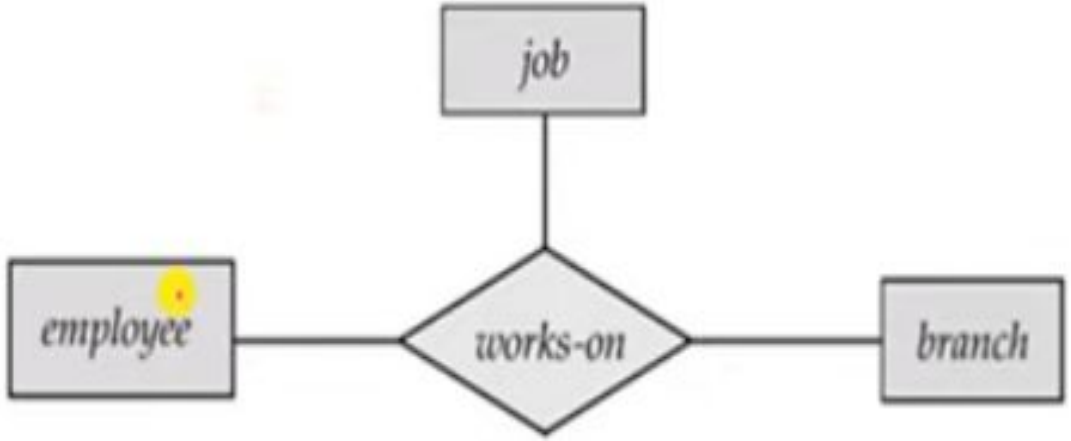
Specialization

3. Aggregation

- **Aggregation** is a process in which a single entity alone is not able to make sense in a relationship so the relationship of two entities acts as one entity.

Consider a ternary relationship **works_on** between **Employee, Branch and Job.**

- Am **employee** ***works on*** a particular **job** at a particular **branch**



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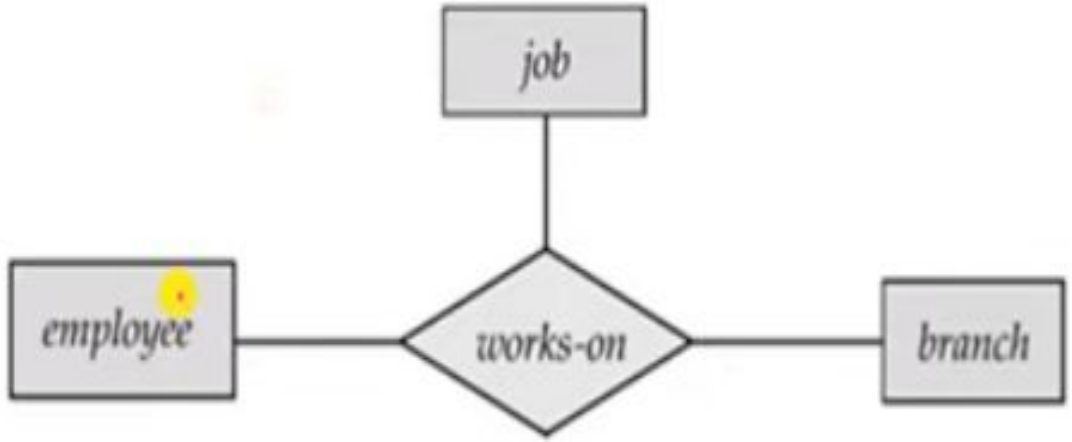
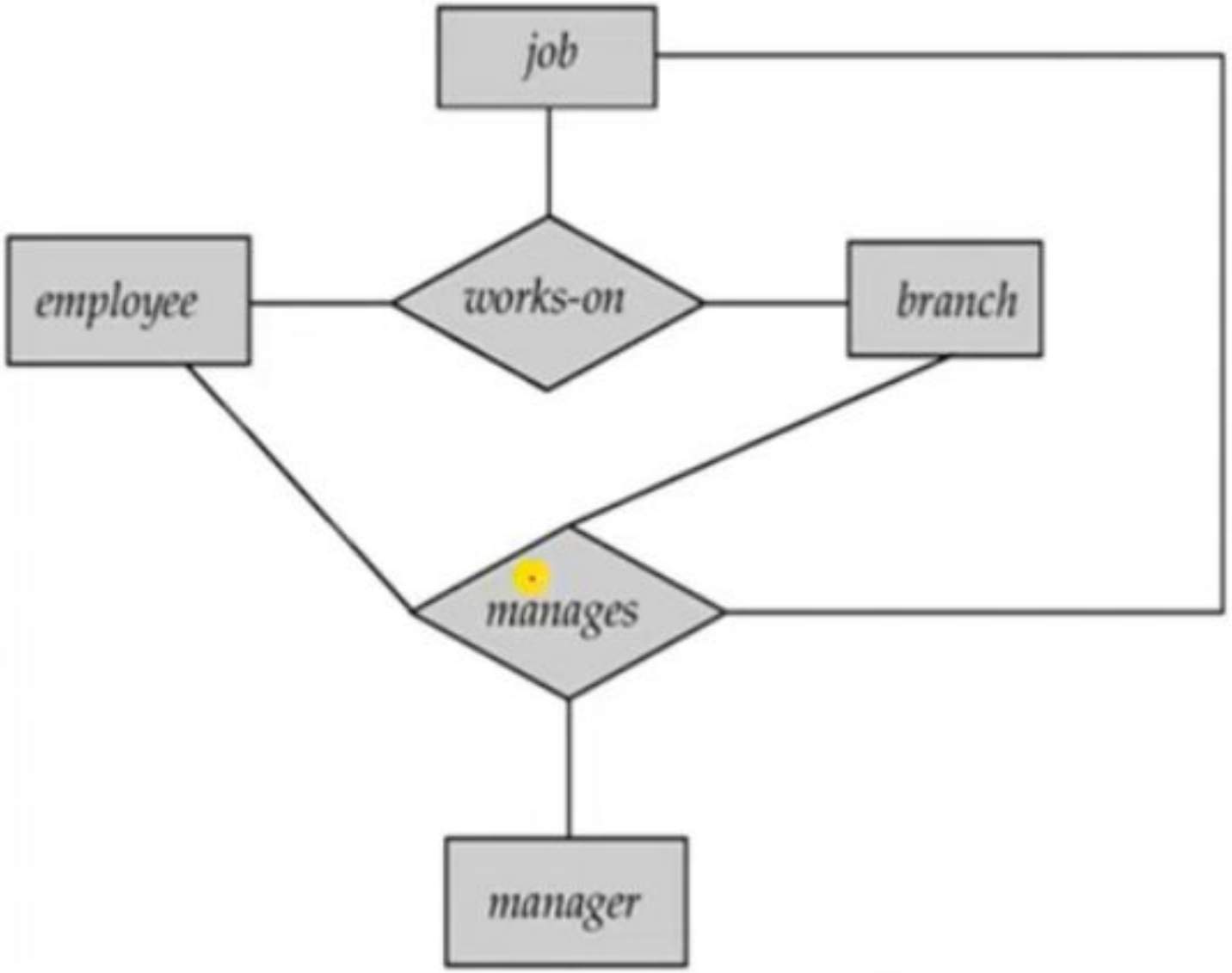
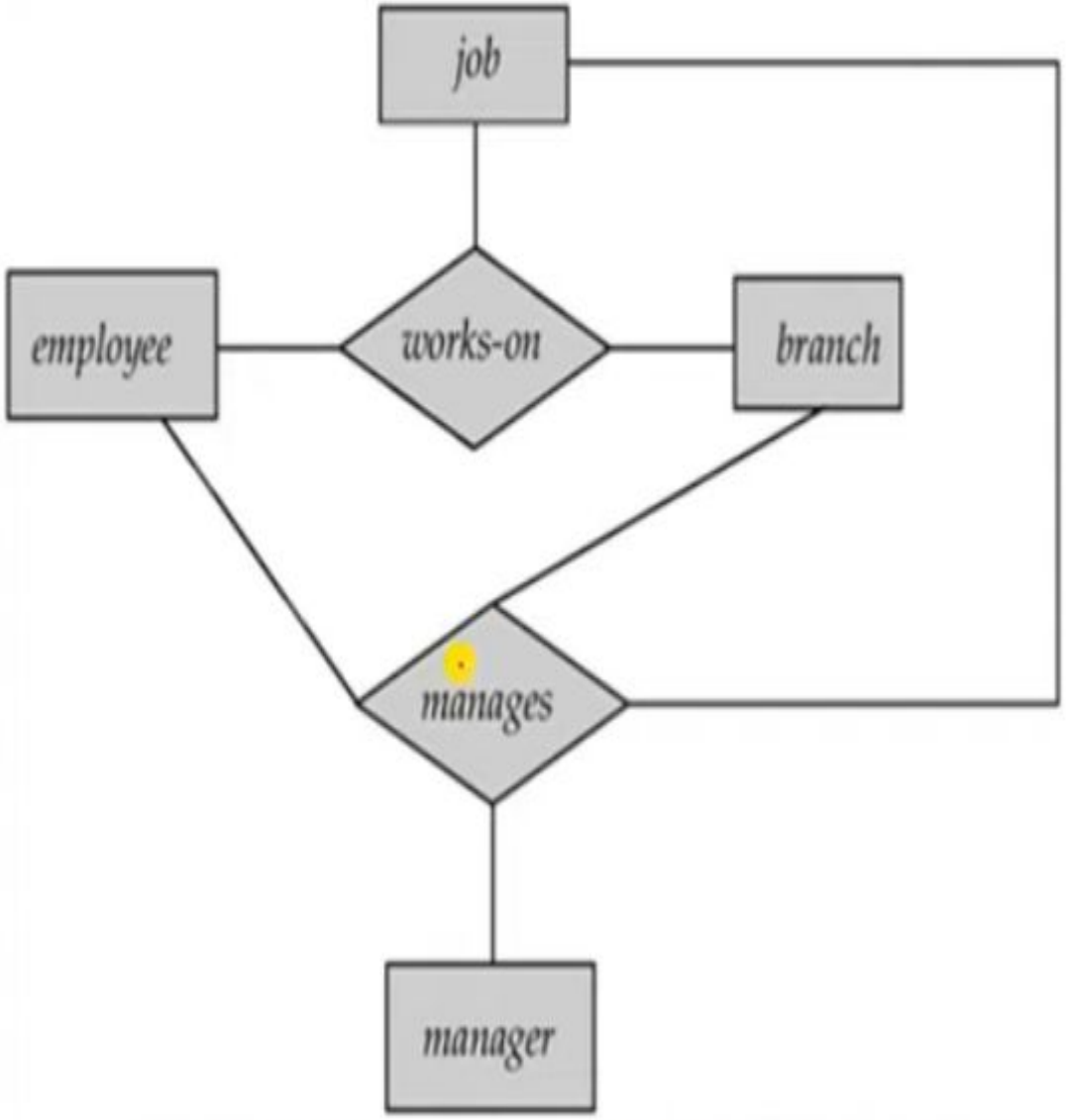


Suppose we want to assign a **manager** for jobs performed by an employee at a branch (i.e. want to assign managers to each employee, job, branch combination)

- Need a separate manager entity-set
- Relationship between each manager, employee, branch, and job entity



ER Diagram with Redundant Relationship

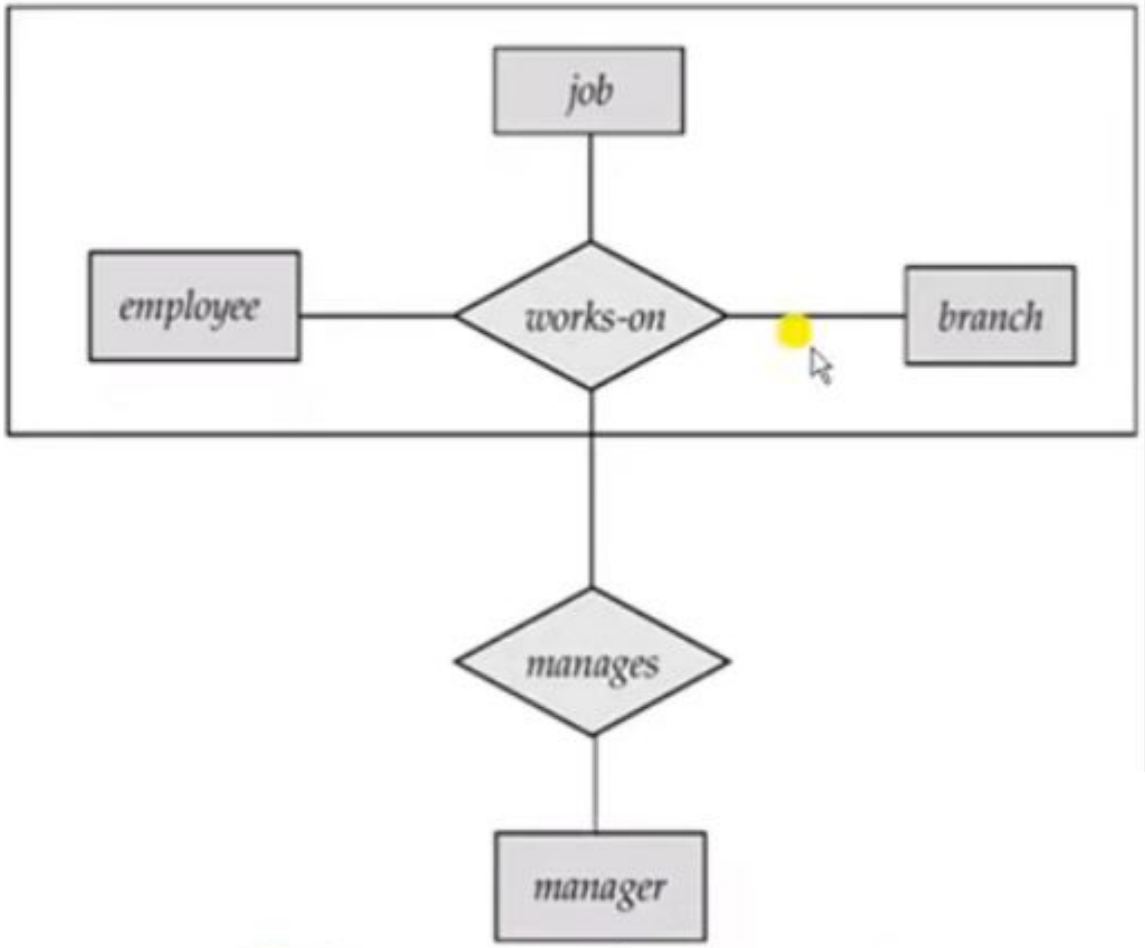


ER Diagram with Redundant Relationship

- Relationship sets ***works-on*** and ***manages*** represent overlapping (redundant) information
  - Every *manages* relationship corresponds to a *works-on* relationship
  - However, some *works-on* relationships may not correspond to any *manages* relationships
    - So we can't discard the *works-o* relationship

Example: (Aggregation)

- With **Aggregation** (without introducing redundancy) the ER diagram can be represented as:
  - An employee works on a particular job at a particular branch
  - An employee, branch, job combination may have an associated manager



ER Diagram with Aggregation